Hayeon Lee

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Research Interest

Robot learning and Vision-Language-Action (VLA) models, with interests in multimodal AI for robotics and vision-based robot control.

Education

Gachon University (Gyeonggi-do, South Korea)

B.S., Artificial Intelligence — Mar. 2024 - Present

- Cumulative GPA: 4.19 / 4.5 (Class rank: 31 / 235)
- Academic Excellence Scholarship, National Program of Excellence in Software (Sept. 2024)
- Credit Exchange Program, Seoul National University
- Silicon Valley AI/AX Technology Program, San Jose State University, USA (July 2026; one of 5 selected students, fully funded)

Research Experience

- **Undergraduate Research Intern**, **Seoul National University**, Seoul — Feb. 2026 - Present
  - Advisor: Dr. Namjoon Kim (Research Professor)
  - Participating in undergraduate research on robotics, multimodal AI, and embedded AI systems.
  - Studying robot learning and vision-language-action models, and AI model applications in robotics environments.
- Undergraduate Research Assistant, **ISSR Lab, Gachon University**, Gyeonggi-do — May 2025 - Nov. 2025
  - Participated in undergraduate research on the Python debugger and CPython debugging environment.

Publications

In conference proceedings

1. Analysis of GDB-based CPython Debugging Environment
   Co-author
   KIPS 2025 (Korea Information Processing Society Conference, Paper ID KIPS_C2025B0232)

Projects

- ROS2–Gazebo Autonomous Robot Mission, **Seoul National University** (course project) Link — Fall 2025
  - Course project for Building Robot Artificial Intelligence, taken through the Credit Exchange Program at SNU.
  - Built a robot mission system by integrating perception, planning, and localization nodes in ROS2 and Gazebo.
  - Responsible for object perception and YOLO fine-tuning; participated in final node integration and mission code development.
  - Core technology: ROS2, Gazebo, YOLO fine-tuning, object detection, path planning.

Teaching Experience

Teaching Assistant, **AI-Centered World Program**, Gachon University — Sept. 2025 - Dec. 2025, Mar. 2026 - June 2026

Teaching Assistant, **Introduction to AI Programming**, Gachon University — Mar. 2026 - June 2026

Technical Skills

- **Programming:** Python, Java, C, Dart, Git, Linux/Ubuntu
- **AI / ML:** PyTorch, TensorFlow, NumPy, pandas, CNN / Transformer, model training and evaluation, robot policy models, VLA models
- **Robotics / Simulation:** ROS2, Gazebo, LeRobot, RoboCasa, robot learning, vision-based robot control
- **Computer Vision:** OpenCV, YOLO fine-tuning, U-Net

Relevant Coursework

- **AI & Robotics:** Computer Vision (A+), Advanced Topics in Artificial Intelligence (A+), Introduction to AI Programming (A+), Introduction to Artificial Intelligence (A, SNU), How to Make a Robot with Artificial Intelligence (A, SNU), Robotics (A), AI-Centric World (A)
- **CS Core:** Problem Solving Methods (A+), Object-Oriented Programming (A+), Computer Network & Programming (A+), Data Structures (A), Operating System (A)

- **Mathematics:** AI Mathematics (A+), Software Mathematics (A+), Probability & Statistics (A+)
- **Research Writing:** English Research Paper Writing for Graduate Students (P, SNU)
- **In Progress (Fall 2026):** Automatic Control Systems, Algorithms